

KINNECT

Local Campus Activity Coordination Platform

User Manual | Developer Setup Guide | SRS

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PART 1 - SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1.1 Purpose & Scope

Kinnect is a hyperlocal, real-time campus platform that helps students discover, create, and coordinate activities (called "Floats"). It bridges the gap between digital communication and physical meetups using proximity-based feeds, live mapping, and instant group chat - all wrapped in a premium glassmorphic mobile-first interface.

Core Goal

Enable students to spontaneously coordinate campus activities in real time, with zero friction - from discovering a study group to joining a football game.

Provide safety mechanisms (SOS, moderation) that make the campus feel connected and secure.

1.2 Tech Stack & Libraries Required

Frontend (Client-Side)

Framework	Next.js 14 with React 18 (Pages Router)
Styling	Tailwind CSS with clsx and tailwind-merge
Maps	Leaflet & React-Leaflet (OpenStreetMap tiles)
Animations	Framer Motion (spring physics)
Authentication	NextAuth.js - Google OAuth 2.0 provider
Data Fetching	SWR (Stale-While-Revalidate) - 5s polling interval
Icons	Lucide-React
Audio/Haptics	Web Audio API + Vibration API

Backend (Server-Side)

Framework	FastAPI (Python) - Async, high-performance
Database	SQLite stored in kinnect.db
ORM	SQLAlchemy (Async mode)

Server	Uvicorn ASGI server
Real-time	FastAPI WebSockets - chat.py (group), dm.py (DMs)
Tunneling	Cloudflare Tunnels for cross-device local testing

1.3 System Architecture Constraints

- Client Architecture: PWA-compatible, mobile-first. Desktop capped at max-w-md.
- Communication: REST APIs for all CRUD; WebSockets for real-time chat and SOS.
- Security: OAuth 2.0 authentication; JWT-based stateless token validation on all API endpoints.
- Proximity: All distance calculations use the Haversine formula on the server side.
- Offline: Personal reminders stored in localStorage - no backend round-trip required.
- Deployment: Next.js rewrites proxy /api/backend/* to FastAPI, unifying the domain and eliminating CORS issues.

PART 2 - DEVELOPER SETUP GUIDE

2.1 Prerequisites

Ensure the following are installed on your system before proceeding:

Node.js	v18 or higher - https://nodejs.org
Python	v3.10 or higher - https://python.org
Git	Latest version - https://git-scm.com

Before You Begin

Clone the repository: `git clone <repo-url>`

Open two separate terminal windows - one for the backend, one for the frontend.

Always start the backend FIRST before the frontend.

2.2 Backend Setup

Open Terminal 1 and run the following commands in order:

1**Navigate to the backend directory**`cd backend`**2****Create a Python virtual environment**`python -m venv venv`**3****Activate the virtual environment**`Windows: venv\Scripts\activate | macOS/Linux: source venv/bin/activate`**4****Install Python dependencies**`pip install -r requirements.txt`**5****Run database migrations**`python migrate_db.py`

6

Start the FastAPI server`uvicorn src.main:app --reload --host 0.0.0.0 --port 8000`**Backend Running**

The backend API is now live at: <http://localhost:8000>

Interactive API docs available at: <http://localhost:8000/docs>

Leave this terminal running while using the app.

2.3 Frontend Setup & Environment Variables

Open Terminal 2 (keep Terminal 1 running) and run:

1

Navigate to the frontend directory`cd frontend`

2

Install all Node modules`npm install`

Configure Google OAuth

Create a file named `.env.local` inside the `frontend/` directory with the following content:

```
# frontend/.env.local
GOOGLE_CLIENT_ID=your_google_cloud_client_id
GOOGLE_CLIENT_SECRET=your_google_cloud_client_secret
NEXTAUTH_SECRET=a_random_secure_string
NEXTAUTH_URL=http://localhost:3000
```

Getting OAuth Credentials

Go to <https://console.cloud.google.com> and create a new project.

Enable the Google OAuth 2.0 API.

Under Credentials, create an OAuth Client ID (Web Application type).

Add `http://localhost:3000/api/auth/callback/google` as an Authorized Redirect URI.

Copy the Client ID and Secret into .env.local.

2.4 Running the Application

1

Start the Next.js development server

`npm run dev`

Application Ready

Frontend is live at: `http://localhost:3000` - open in your browser.

Backend API running at: `http://localhost:8000`

Both servers must be running simultaneously for the app to work.

PART 3 - END-USER MANUAL

3.1 System Requirements

Before using Kinnect, ensure your device meets the following requirements:

Browser	Google Chrome, Apple Safari, or Mozilla Firefox (latest version)
Location Services	Must be ENABLED - the proximity feed requires GPS access
Account	A valid Google Account is required to sign in
Connection	Active internet connection required for live feed and chat
Install as PWA	Optional: tap 'Add to Home Screen' for a native app experience

Location Permission

When you first open Kinnect, your browser will ask for Location permission.

You MUST tap 'Allow' - without it, the proximity feed cannot calculate distances.

Kinnect only uses your location to filter nearby activities. It is never stored permanently.

3.2 Getting Started - Sign In & Onboarding

1**Open Kinnect in your browser**

Navigate to the Kinnect URL or open it from your Home Screen.

2**Sign in with Google**

Tap the 'Sign in with Google' button and select your Google account.

3**Complete your profile (first-time only)**

Select your Hall of Residence from the dropdown. Type your Interests separated by commas (e.g., coding, basketball, music). Tap Complete Profile.

4**You're in!**

You'll be taken to your dashboard. The live feed will automatically load nearby activities.

Tip - Profile Completion

Your Hall of Residence and Interests help other users find you as a potential activity partner.

You can update your profile at any time by tapping your avatar in the navigation dock.

3.3 Discovering Activities (Floats)

Activities on Kinnect are called Floats. There are three ways to discover them:



The Dashboard Feed

Your main screen shows a live feed of Floats near you, sorted by distance. Each card shows the activity type, distance, capacity bar (green → amber → red as it fills), time, and join mode. The feed refreshes automatically every 5 seconds.



The Map View

Tap the Map icon in the navigation dock to see all active Floats as emoji pins on a full-screen campus map. Tap any pin to see a preview of that activity.



Filtering the Feed

Tap the Filter button to narrow results by: Maximum distance radius, Date and time of day, Join mode (Open vs. Approval-only). The feed updates instantly as you adjust filters.

Understanding Activity Cards

Capacity Bar	Green = plenty of space Amber = filling up Red = almost full Full = locked
Distance	Real-time straight-line distance from your current location
Tier Badge	Hall (Tier 1) = within your residence Hub (Tier 2) = nearby General (Tier 3) = campus-wide
Join Mode	Open = join instantly  Approval = host must accept your request
Expiry	Activities disappear from the feed after their expiry time passes

3.4 Creating a Float

You can host up to 5 active Floats at a time. Here's how to create one:

1**Tap the glowing '+' button**

This is the floating action button (FAB) on your dashboard.

2**Fill in the details**

Title - a short, descriptive name. Description - tell people what this Float is about. Category - pick an emoji that represents the activity type. Tier - Hall, Hub, or General campus.

3**Pick a location**

An interactive map will appear. Tap the exact spot where the activity will happen to drop a pin.

4**Set capacity and time**

Max Capacity - the maximum number of participants allowed. Event Time - when the activity starts. Expiry Time - when the Float disappears from the feed.

5**Choose a Join Mode**

Open: Anyone can join instantly until capacity is reached. Approval: Participants must request to join, and you approve or reject each request.

6**Tap 'Create Float'**

Your Float goes live immediately and appears in the feed of nearby users.

Managing Your Float

To disband your Float at any time, open it and tap 'Disband Activity'.

Disbanding removes all participants and permanently closes the group chat.

You can view all your hosted Floats under My Activities in the dashboard.

3.5 Joining Activities & Chat

Joining a Float

Open Mode	Tap an activity card → tap Join. You're added instantly. The activity appears in your Joined list.
Approval Mode	Tap an activity card → tap Request to Join. The host receives a notification to approve or reject your request. You'll be notified of the outcome.
Full Activity	The Join button is disabled and shows 'Full' in red when max capacity is reached.
Leaving	Open the activity → tap Leave Activity. You'll be removed from the participant list and chat.

Real-Time Group Chat

Every Float has a dedicated group chat that is automatically created when the activity goes live.



Opening the Chat

Tap an activity you've joined, then tap the Chat icon to open the LiveChatDrawer. All historical messages are loaded instantly.



Unread Badge

A red pulsing badge appears on the Chat icon in your navigation dock whenever you have unread messages in any activity chat.



Auto-Close on Disband

If the host disbands the activity, the chat drawer closes automatically and the conversation history is preserved in your Chats tab.



Chat History

Even after an activity expires and leaves the feed, you can still access the chat from the Chats tab until you explicitly delete it.

3.6 Friends & Direct Messages

1

View a user's profile

Tap on any participant's avatar inside an activity chat or on an activity card to open their profile.

2

Send a Friend Request

Tap 'Add Friend' on their profile. They'll receive a notification. The button will show 'Pending' until they respond.

3

Accept or Decline

Incoming friend requests appear in your Notification Center. Tap Accept or Decline.

4

Start a Direct Message

Once you're mutual friends, tap 'Message' on their profile or open the DMs tab to start a private conversation.

DM Privacy

Direct Messages are only available between mutual friends (both users must have accepted each other).

DM conversations persist indefinitely until one party deletes them.

3.7 Emergency SOS System

EMERGENCY USE ONLY

The SOS feature is intended for genuine campus emergencies only.

Misuse will result in account penalties. See Section 3.8 for details.

If you are in a genuine emergency situation on campus, follow these steps:

1**Tap the red SOS button in the header**

The button is always visible at the top of the app. Tap it to open the Emergency Modal.

2**Confirm the alert**

You must confirm the SOS to prevent accidental triggers. This captures your exact GPS coordinates.

3**Alert is broadcast**

A flashing red banner appears on ALL active users' screens campus-wide. Your location is visible on the map as a pulsing red marker.

4**Responders arrive**

Nearby users see your location and can tap 'Respond' to let you know they're on their way.

5**Resolve the emergency**

Once the situation is handled, you or a responder marks the emergency as Resolved and classifies it as a true alarm or false alarm.

SOS False Alarm Policy

Strike 1	A warning notification is sent to your account.
Strike 2	Your account is suspended for 10 days (banned_until = now + 10 days).
During Ban	You cannot create Floats, join activities, or send messages until the ban expires.

3.8 Community Guidelines & Bans

Kinnect relies on community trust. Please use all features responsibly.

Reporting Inappropriate Activities

- Tap the ... menu on any activity card → tap Report Activity.
- Reports are anonymous. The activity creator is not notified of individual reports.
- If an activity receives 5 or more reports, it is automatically disbanded and the creator is banned for 5 days.

Ban Summary

Activity Auto-Ban	5 reports on your activity → 5-day account ban
SOS False Alarms	2 false SOS alarms → 10-day account ban
During a Ban	Login shows a ban message with the expiry date. All features are locked.
Ban Expiry	Bans lift automatically when the timestamp passes. No action needed.

3.9 Personal Organizer & Reminders



Calendar View

Tap the Calendar icon in your dashboard to open the Personal Organizer. Dates with joined or hosted activities show a purple dot. Dates with personal reminders show a green dot.



Adding Reminders

Inside the calendar, tap any date and then tap 'Add Reminder'. Type your reminder text and set a time. Reminders are saved instantly to your device (offline-capable).



My Activities

Tap the My Activities button to see all Floats split into 'Hosted by me' and 'Joined by me', giving you a quick overview of your current commitments.

Offline Reminders

Personal reminders are stored locally on your device using localStorage.

They are available even without an internet connection.

Reminders are device-specific - they do not sync across multiple devices.

3.10 Notifications

The Notification Center (bell icon in the dock) tracks all important events on your account:

Join Requests	Someone has requested to join your activity - tap to Approve or Reject.
Request Approved	Your join request was accepted. Tap to open the activity and chat.
Request Rejected	Your join request was declined.
Activity Disbanded	An activity you were part of has been disbanded by the host.
Friend Request	Someone wants to connect with you - tap to Accept or Decline.
Friend Accepted	A user accepted your friend request. You can now DM them.
Chat Message	Consolidated chat alert: 'Someone texted you in [Activity Name]'.
SOS False Alarm	A warning if your SOS was classified as a false alarm.

Smart Consolidation

Chat notifications are consolidated - if multiple messages arrive in the same activity, you see one notification with an updated timestamp instead of dozens.

A red pulsing badge on the Chat dock icon means you have unread chat messages.

A numbered badge on the Bell icon means you have unread non-chat notifications.

QUICK REFERENCE CARD

Keyboard / Gesture Shortcuts

Tap FAB (+)	Create a new Float
Tap activity card	View activity details
Long-press SOS button	Open Emergency Modal
Swipe down on drawer	Close any bottom sheet drawer
Tap avatar	View your profile or another user's profile
Tap map pin	Preview that activity on the map

Glossary

Float	A Kinnect activity broadcast - the core unit of the platform
FAB	Floating Action Button - the glowing '+' that creates a Float
Join Mode	Open (instant) or Approval (host-gated) - set per Float
Tier	Activity scope: Tier 1 = Hall, Tier 2 = Hub, Tier 3 = General
SWR	Stale-While-Revalidate - the feed's background refresh mechanism
Haversine	Distance formula used to calculate proximity between two GPS points
SOSAlertBanner	The full-screen flashing red banner shown during an emergency
LiveChatDrawer	The sliding chat panel attached to each Float
PWA	Progressive Web App - install Kinnect directly to your Home Screen
DM	Direct Message - private 1-on-1 chat between mutual friends